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REVIEW

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A PRISMA systematic review through time on predictive musculoskeletal simulations

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Abstract

This PRISMA systematic review covers the literature on predictive, musculoskeletal simulations. First, we define predictive movement for musculoskeletal systems, as the current literature suffers from inconsistent nomenclature. We distinguish two methods of prediction. The first uses neural models, like muscle-reflex-based and central pattern generator models. The second uses optimization, to make up for the lack of a neural model, like optimal control and deep reinforcement learning. For each method, we illustrate the main concepts and report accuracies, simulation times and limitations. Moreover, we identified key works over the past 50 years, which are fundamental for the current state-of-the-art. The majority of works employ optimization. We recognize six classes of cost function terms and note they are often combined using linear combinations. We describe musculoskeletal models, their muscle model, ground contact model and personalization. Similarly, we identify key software like OpenSim and SCONE. Additionally, we provide an overview of simulated movements, pathologies and assistive devices. We emphasize the difference in tracking simulations and prediction, while clarifying the benefits of using experimental data to predict movement. Finally, we call for quantitative validation to establish comprehensive comparisons between methods. To this end, we share a list of works open-sourcing their codes.

Keywords Gait, Human, Locomotion, Musculoskeletal simulations, Optimization, Predictive, Reinforcement learning, Central pattern generators, Muscle-reflexes

Background

Musculoskeletal simulations use virtual models of the human body to study different types of motion, like walking [1], jumping [2], reaching [3] and more. Typically, musculoskeletal (MSK) models are driven by experimental data, like body kinematics [4, 5], ground reaction forces or electromyography (EMG) signals [6]. However, predictive simulations, that can extend beyond available experimental data, are required to predict the effects of assistive devices, surgeries and pathologies, and to understand the human neural controller [7]. A clear definition of simulations, predicting kinematics and kinetics of the MSK system, is still missing in the literature.

Since 1970, researchers have developed different approaches to predictive simulations, starting with Hatze et al. [8] using optimization to move a model of the right leg from an initial to a final point in the shortest time possible. The first objective of this review (RO1) is to overview existing methods to predict the movement and kinetics of MSK systems. Secondly (RO2), we create a timeline of key historical developments.

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